

ITA0608 - MACHINE LEARNING

LAB PROGRAMS

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Experiment 13: Car Price Prediction Model using Python

Aim

To implement a Car Price Prediction model using Linear Regression in Python and predict the price of a car based on its features.

Procedure

1. Import the required libraries.
2. Create a car dataset containing features such as year, mileage, engine size, and car age.
3. Separate input features and target price.
4. Split the dataset into training and testing data.
5. Train the Linear Regression model.
6. Predict car prices using the test data.
7. Evaluate the model using MSE and R^2 score.
8. Predict the price of a new car.

Algorithm

1. Start.
2. Load the car dataset.
3. Select the input features and target variable.
4. Split the dataset into training and testing sets.
5. Train the Linear Regression model.
6. Predict prices for the test data.
7. Calculate MSE and R^2 score.
8. Use the trained model to predict the price of a new car.
9. Display the results.

Code

```
import numpy as np
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score

X = np.array([
    [2018, 25000, 1.2],
    [2019, 20000, 1.5],
    [2017, 35000, 1.0],
    [2020, 15000, 1.5],
    [2021, 10000, 2.0],
    [2016, 50000, 1.2],
    [2019, 30000, 1.5],
    [2022, 8000, 2.0],
    [2015, 60000, 1.0],
    [2020, 18000, 1.8],
    [2018, 40000, 1.5],
    [2021, 12000, 2.0]
])

y = np.array([
    750000, 850000, 600000, 1050000,
    1300000, 500000, 780000, 1450000,
    400000, 1100000, 650000, 1250000
])

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

model = LinearRegression()

model.fit(X_train, y_train)

y_pred = model.predict(X_test)

print("Actual Prices:")
print(y_test)

print("Predicted Prices:")
print(y_pred)

print("Mean Squared Error:", mean_squared_error(y_test, y_pred))

print("R2 Score:", r2_score(y_test, y_pred))

new_car = [[2022, 10000, 1.5]]

prediction = model.predict(new_car)

print("Predicted Car Price:", prediction[0])
```

Output

```
... Actual Prices:
[ 650000 1100000  750000]
Predicted Prices:
[ 807268.60299599 1103938.52165797  723951.99209384]
Mean Squared Error: 8475000038.454495
R2 Score: 0.7722917243400285
Predicted Car Price: 1247725.054045409
```

Result

The Car Price Prediction Model was successfully implemented using Linear Regression. The model predicts the price of a car based on its manufacturing year, mileage, and engine size and evaluates its performance using MSE and R^2 score.

